EPPS 6302 Methods of Data Collection and Production: Assignment 6

Text analytics with quanteda

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> **What you will produce**
>
> Three corpora read with quanteda — archived summit tweets, presidential documents you collect from GovInfo, and the inaugural addresses — and a short report on what each kind of text can and cannot tell you, including what Wordfish measures.

0. Set up

a. Install once, in the Console: `install.packages(c("quanteda", "quanteda.textstats", "quanteda.textplots", "quanteda.textmodels", "tidyverse"))`
b. Download `quanteda_textanalytics01.R` from Teams. Parts 1–3 below follow its three parts.
c. You need the GovInfo key and the functions from `govtdata01.R` (Assignment 5) for Part 2.

1. Read

a. *quanteda* — https://quanteda.io, and the tutorials at https://tutorials.quanteda.io, ch. 1–4 and 6.

2. Analyze

a. One summit, two kinds of text. In November 2021 Presidents Biden and Xi met by video.

i. **The public** (Part 1). Tweets collected during the summit: frequent words, hashtag and mention networks, LSA. This dataset is **archived** — the API that produced it no longer offers that access, so it cannot be collected again.
ii. **The government** (Part 2). Collect every document in the *Compilation of Presidential Documents* (GovInfo, `CPD`) mentioning "Xi Jinping" since 20 January 2021. Read Xi in context with `kwic()`, then use keyness to compare the Biden and Trump administrations.
iii. Compare: what does each source let you say about the summit, and about US–China relations? Who produced each text, and for whom?
iv. *Optional.* For a public counterpart in 2025, collect YouTube comments on news coverage of the Trump–Xi meeting in Busan (30 October 2025) with your YouTube key, and compare with (i).

b. US presidential inaugural addresses (Part 3), 1945 onward.

i. Any **similarities and differences** over time and among presidents? Use length, lexical diversity, readability, cosine similarity, and keyness by party.
ii. Check which is the latest address in `data_corpus_inaugural`. If the 2025 address is missing, say how you would add it — and from which source.

3. What is Wordfish?

a. Research it on the *quanteda* site (`textmodel_wordfish()`) and in tutorial ch. 6; *rec.* Slapin & Proksch (2008).
b. Run it on the inaugural addresses (Part 3d). In a paragraph: what does the model assume about how words are generated? What does the dimension it found **seem** to measure, and what would you need to show that it **does**? Why must you set `dir`?

> **Every corpus has a data-generating process.** The tweets were chosen by a keyword and a collection window; the presidential documents by an editor at the Office of the Federal Register and by your query; the inaugurals by a tradition. Say which in your report — it decides what your keyness and Wordfish results can mean.

4. Submit

a. Publish as `assign06.qmd` under **Assignments** on your website, with the plots, a short report (about 600 words), and the data note for the `CPD` collection (source, permission, query, selection, gaps). Send the URL to the teaching assistant if you have yet to. Post the log and manifest — **not** the documents, and not your key.